SUMS OF TWO SQUARES WITH FEW PRIME FACTORS, II: THE MEAN SQUARE OF THE CUSPIDAL ERROR OF $\theta^2(mz)\theta^2(m'z)$, ALMOST ALL N, AND THE SIZE OF THE PETERSSON NORM

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ABSTRACT. Let m, m' be coprime odd integers, $\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z)$, $G_{m,m'}$ its cuspidal part and $L = 4mm'$, so that $a_G(N) = T(m, m'; N) - M(m, m'; N)$ is the error term in the number of representations of N by $m(x^2+y^2)+m'(z^2+w^2)$. Part I proved $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$ for the Petersson norm and deduced the level of distribution $1/2$ for the weighted sequence $r(n)r(N-n)$ and the bounds $\Omega(n(N-n)) \le 8$, $\max(\Omega(n), \Omega(N-n)) \le 5$ for every large $N \equiv 2 \pmod 4$. Here we prove a mean-square bound $\sum_{N\le Y} |a_G(N)|^2 \le 4\pi e^2 Y(1 + 54Y/L)\|G\|^2$ for every $G \in S_2(\Gamma_0(L))$ by a covering argument; for $G_{m,m'}$ it says that the root mean square of $(T - M)/\sqrt{N}$ is $\ll (mm')^{-1/2+\varepsilon}$, and it gives the level of distribution $1 - \varepsilon$ for the weighted sequence for all but $O(X^{1-\delta})$ integers $N \le X$, and for those N the bounds 5 and 4 (a result which, as an arithmetic statement about almost all N, is superseded by the circle method, which gives $N = p + q$ with $p \equiv q \equiv 1 \pmod 4$). We reformulate the uniform bound of Part I in terms of Poincaré series, state the individual bound $|T - M| \ll N^{1/2+\varepsilon}(mm')^{-1/2+\varepsilon}$ suggested by the data, and show that any power saving in mm' for individual N raises the level of distribution for every N beyond $1/2$. Numerically, $\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2$, with an exact check at the level 20, and the supremum of $|T - M|/\sqrt{N}$ over N exceeds the root mean square by a bounded factor at every level computed.

1. RESULTS

We keep the notation of [2], referred to as Part I. Thus $r(n)$ is the number of representations of n as a sum of two squares, m, m' are coprime odd positive integers,

$$T(m, m'; N) = \sum_{ma+m'b=N} r(a)r(b), \qquad L = 4mm',$$

$$\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z) = \sum_{N\ge 0} T(m, m'; N)e(Nz) \in M_2(\Gamma_0(L)),$$

$\Theta_{m,m'} = E + G$ is the decomposition into the Eisenstein part, whose coefficients are the Siegel main terms $M(m, m'; N)$, and the cuspidal part $G = G_{m,m'} \in S_2(\Gamma_0(L))$, so that

$$a_G(N) = T(m, m'; N) - M(m, m'; N) \qquad \text{for every } N \ge 1.$$

The Petersson norm is $\|G\|^2 = \int_{\Gamma_0(L)\backslash\mathbb{H}} |G|^2 \,\mathrm{d}x\,\mathrm{d}y$. Lemma 2.1 of Part I, which is Waibel's theorem [6, Theorem 1] for the form $Q_{m,m'} = m(x^2+y^2)+m'(z^2+w^2)$, gives $\|G\| \ll (mm')^{1/4+\varepsilon}$, and the first two versions of Part I deduced from it, through a Hecke operator, the uniform bound $a_G(N) \ll d(N)N^{1/2}(mm')^{1/4+\varepsilon}$ for $(N, mm') = 1$, $v_2(N) \le 1$, the level of distribution $2/5$ and the bounds 9 and 6, and conjectured (Conjecture 5.1 there) that the factor $(mm')^{1/4}$ can be removed. Version 3 of Part I, which this paper accompanies, proves that the norm itself is much smaller than Waibel's general bound, by computing the expansion of $\Theta_{m,m'}$ at every cusp of $\Gamma_0(L)$ with Waibel's method, and draws the consequences. We recall what we use.

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Theorem 1.1 (Part I, Theorems 1.2 and 2.3)**.** *Let m, m' be coprime odd positive integers.*

(i) $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$.

(ii) *For $N \geq 1$ with $(N, mm') = 1$ and $v_2(N) \leq 1$,*

$$T(m, m'; N) = M(m, m'; N) + O_\varepsilon\big(d(N)\, N^{1/2} (mm')^\varepsilon\big);$$

for cube-free m, m' and every N with $v_2(N) \leq 1$ the error term is $O_\varepsilon(d(N)^4 N^{1/2}(mm')^\varepsilon)$ (Part I, Corollary 2.7).

Part I deduces from (ii) that $r(n)r(N-n)$ has level of distribution $1/2$ (Theorem 3.2 there) and, with sieve parameters certified in interval arithmetic, that for every sufficiently large $N \equiv 2 \pmod 4$

$$\#\big\{1 \leq n < N :\ n = x^2 + y^2,\ N - n = z^2 + w^2,\ \Omega(n(N-n)) \leq 8,\ \max(\Omega(n), \Omega(N-n)) \leq 5\big\} \geq\ 0.23\, \mathfrak{S}_r(N)\, \frac{N}{(\log N)^2}, \qquad \mathfrak{S}_r(N) = \prod_{\substack{p \mid N \\ p > 2}} \frac{p-1}{p-2},$$

(Theorem 1.1 there; versions 1–2 had 9 and 6). The proof of (i) shows that the expansion of $\Theta_{m,m'}$ at every cusp is again a theta series of a form $A(x_1^2 + x_2^2) + B(x_3^2 + x_4^2)$ with $AB \asymp mm'$, whose coefficients are $O(n^\varepsilon)$ for $n \ll mm'$ because $A\alpha + B\beta = n$ has $O(1)$ solutions, and suggests $\|G_{m,m'}\|^2 \asymp (\log mm')^2$ (Part I, Remark 2.13); numerically, $\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2$ (Section 5).

The first theme of this paper is the average over N of the error term, which is of independent interest and gives a stronger level of distribution for almost all N.

Theorem 1.2 (Mean square)**.** *Let $L \geq 1$, $G \in S_2(\Gamma_0(L))$ and $Y \geq 1$. Then*

$$\sum_{n \leq Y} |a_G(n)|^2 \ \leq\ 4\pi e^2\, Y \Big(1 + \frac{54\, Y}{L}\Big) \|G\|^2. \tag{1}$$

In particular, for coprime odd m, m', $G = G_{m,m'}$ and $Y \geq mm'$, by Theorem 1.1(i),

$$\sum_{N \leq Y} \big|T(m, m'; N) - M(m, m'; N)\big|^2 \ \ll_\varepsilon\ Y^2\, (mm')^{-1+\varepsilon}, \tag{2}$$

and hence, since $\sum_{Y/2 < N \leq Y} |a_G(N)|^2/N \ \leq\ (2/Y) \sum_{N \leq Y} |a_G(N)|^2$, the root mean square of $(T - M)/\sqrt{N}$ over $Y/2 < N \leq Y$ is $\ll (mm')^{-1/2+\varepsilon}$.

The proof of (1) is a classical covering argument and does not use the Petersson formula; the dependence on the level is exactly the factor Y/L, i.e. the volume of $\Gamma_0(L)\backslash\mathbb{H}$. The bound (2) is the exact order of magnitude: by the Rankin–Selberg method the mean square of $(T - M)/\sqrt{N}$ is asymptotic to $16\pi^2 \|G\|^2 / \operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})$, and Table 1 shows the data.

Averaging over N is natural for the arithmetic application, since the sieve of Part I is applied to one N at a time and an exceptional set of N can be tolerated. Let, as in Part I, $N \equiv 2 \pmod 4$, $\mathcal{A} = \{n \leq N :\ (n, N) = 1\}$, $w(n) = r(n)r(N-n)$, and for coprime square-free d, e with $(de, 2N) = 1$ let $R(d, e) = R(d, e; N)$ be the remainder in

$$\sum_{\substack{n \in \mathcal{A} \\ d \mid n,\ e \mid N-n}} w(n) = X_N\, h(d) h(e) + R(d, e; N)$$

(this is $S(d, e) = M(d, e) + R(d, e)$ of Part I, Section 3; we write X_N for the main term X there, to avoid a clash with the length X of the range of N below).

Theorem 1.3 (Level of distribution $1-\varepsilon$ for almost all N)**.** *Let $0<\varepsilon<1/20$, $X\ge X_0(\varepsilon)$ and $D=X^{1-3\varepsilon}$. Then*

$$\sum_{\substack{X/2<N\le X\\ N\equiv 2\ (4)}}\ \sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \ll_\varepsilon\ X^{2-\varepsilon/2}.$$

Consequently, for all but $O_\varepsilon(X^{1-\varepsilon/4})$ integers $N\equiv 2 \pmod 4$ in $(X/2,X]$,

$$\sum_{\substack{de\le N^{1-3\varepsilon}\\ (d,e)=1,\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \le\ N^{1-\varepsilon/5}. \tag{3}$$

The sieve of Part I, Section 4, uses the level of distribution only through (3); running it at the level $\theta=19/20$ gives the following.

Corollary 1.4. *For all but $O(X^{1-\delta})$ integers $N\equiv 2 \pmod 4$ in $(X/2,X]$, with an absolute $\delta>0$,*

$$\#\big\{1\le n<N:\ n=x^2+y^2,\ N-n=z^2+w^2,\ \Omega(n(N-n))\le 5,\ \max(\Omega(n),\Omega(N-n))\le 4\big\} \ge\ 0.34\,\mathfrak{S}_r(N)\,\frac{N}{(\log N)^2},$$

the integers counted being coprime to N and products of distinct primes $\equiv 1 \pmod 4$ exceeding $N^{2/25}$.

The certified sieve parameters are $\alpha=2/25$, $\lambda=1/3$, $\beta_1=69/100$ (Section 3). The pairs $(5,4)$ for almost all N and $(8,5)$ for every N should be compared with $(9,6)$ for every N, which is what Waibel's bound alone gives (Part I, Remark 4.5). A floating-point scan (Figure 2) shows that the best attainable value of $1/\lambda+1/\beta_1$ drops below 4 only for $\theta>0.88$; at $\theta=19/20$ the scan gives 3.83 (at $\alpha=0.10$, $\beta_1=0.47$, $\lambda=0.59$, with $1/\lambda+2/\beta_1=5.96$), i.e. the pair $(5,3)$, which we have not certified, for the reason explained next.

Corollary 1.4 must also be compared with what the circle method gives for almost all N. By the method of Montgomery and Vaughan [4], carried out with the characters modulo 4 (see [1] for Goldbach's problem with primes in prescribed residue classes), all but $O(X^{1-\delta})$ integers $N\equiv 2 \pmod 4$ up to X are sums $p+q$ of two primes $p\equiv q\equiv 1 \pmod 4$, with $\gg \mathfrak{S}_r(N)N/(\log N)^2$ representations; such primes are sums of two squares, so for almost all N the pair $(1,1)$ holds with the count of Corollary 1.4. As an arithmetic statement about almost all N, Corollary 1.4 is therefore far from new. What is new is Theorem 1.3, a level of distribution $1-\varepsilon$ for the weighted sequence $r(n)r(N-n)$, together with the observation that the sieve of Part I extracts $(5,4)$ from it with no further input; the sieve route adds something only for *every* N, where the circle method gives nothing (Goldbach's problem) and where Part I gives $(8,5)$. Section 4 formulates the individual bound that would raise the level to $1-\varepsilon$ for every N and with it give $(5,4)$ for every N (Proposition 4.2).

Section 4 restates the uniform bound of Theorem 1.1(ii) in terms of Poincaré series and formulates the individual bound, and Section 5 contains the numerical evidence: the data of Part I, an exact check of the whole numerical chain at the level 20, the empirical law $\|G_{m,m'}\|^2\approx 0.37(\log mm')^2$, which corrects a remark in the first two versions of Part I, and the supremum of $|T-M|/\sqrt{N}$ over N.

2. Proof of Theorem 1.2

Let $G(z)=\sum_{n\ge1}a(n)e(nz)\in S_2(\Gamma_0(L))$ and put $F(z)=y^2|G(z)|^2$, $z=x+iy$. Since $G(\gamma z)=(cz+d)^2G(z)$ and $\operatorname{Im}(\gamma z)=y/|cz+d|^2$ for $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(L)$, the function F is

$\Gamma_0(L)$-invariant, and with the invariant measure $\mathrm{d}\mu = y^{-2}\mathrm{d}x\,\mathrm{d}y$ we have $\|G\|^2 = \int_{\mathcal{F}} F\,\mathrm{d}\mu$ for any fundamental domain $\mathcal{F}$ of $\Gamma_0(L)$.

Let $y_0 = 1/(4\pi Y)$ and $R = \{x+iy:\ 0 \le x < 1,\ y_0 \le y \le 2y_0\}$ (half-open in x, so that each orbit of the translations $z \mapsto z+b$ meets R at most once). By Parseval, $\int_0^1 |G(x+iy)|^2\,\mathrm{d}x = \sum_n |a(n)|^2 e^{-4\pi n y}$, hence

$$\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \sum_n |a(n)|^2 \int_{y_0}^{2y_0} e^{-4\pi n y}\,\mathrm{d}y \ \ge\ \frac{e^{-2}}{4\pi Y}\sum_{n\le Y}|a(n)|^2,$$

because $4\pi n y \le 8\pi Y y_0 = 2$ for $n \le Y$, $y \le 2y_0$. On the other hand $\int_R |G|^2\,\mathrm{d}x\,\mathrm{d}y = \int_R F\,\mathrm{d}\mu$, and

$$\int_R F\,\mathrm{d}\mu = \sum_{\gamma\in\Gamma_0(L)/\{\pm1\}} \int_{R\cap\gamma\mathcal{F}} F\,\mathrm{d}\mu = \int_{\mathcal{F}} F(w)\,\#\{\gamma:\ \gamma w \in R\}\,\mathrm{d}\mu(w) \ \le\ \mu_R\,\|G\|^2,$$

where $\mu_R = \sup_{z\in R}\#\{\gamma\in\Gamma_0(L)/\{\pm1\}:\ \gamma z\in R\}$ (if $\#\{\gamma:\gamma w\in R\}\neq 0$ for some $w\in\mathcal{F}$, pick γ_0 with $z=\gamma_0 w\in R$; then $\gamma\mapsto\gamma\gamma_0^{-1}$ is a bijection from $\{\gamma:\gamma w\in R\}$ onto $\{\gamma:\gamma z\in R\}$, so the count at w is at most μ_R). It remains to bound μ_R. Fix $z=x+iy\in R$ and let $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(L)$ with $\gamma z\in R$.

- If $c=0$, then $\gamma=\pm\left(\begin{smallmatrix}1&b\\0&1\end{smallmatrix}\right)$ and $\gamma z = z+b$, so $x+b\in[0,1)$ forces $b=0$: one element.
- If $c\neq 0$, we may take $c>0$, since γ and $-\gamma$ are the same element of $\Gamma_0(L)/\{\pm1\}$. Then $L\mid c$, and $\mathrm{Im}(\gamma z) = y/|cz+d|^2 \in [y_0,2y_0]$ together with $y\in[y_0,2y_0]$ gives $|cz+d|^2\in[1/2,2]$. Since $|cz+d|^2 = (cx+d)^2+c^2y^2$, this forces $c^2y^2\le 2$, hence $c\le\sqrt{2}/y_0 = 4\sqrt{2}\,\pi Y$, and $|cx+d|\le\sqrt{2}$. The number of positive multiples c of L with $c\le 4\sqrt{2}\pi Y$ is $\lfloor 4\sqrt{2}\pi Y/L\rfloor \le 4\sqrt{2}\pi\,Y/L$; for each c there are at most 3 integers d with $|cx+d|\le\sqrt{2}$; and for each pair (c,d) the elements of $\Gamma_0(L)$ with bottom row (c,d) are $\left(\begin{smallmatrix}1&k\\0&1\end{smallmatrix}\right)\gamma_0$, $k\in\mathbb{Z}$, whose images $\gamma_0 z+k$ have real part in $[0,1)$ for exactly one k. This gives at most $4\sqrt{2}\pi\cdot 3\,Y/L = 12\sqrt{2}\pi\,Y/L < 54\,Y/L$ elements.

Hence $\mu_R \le 1+54Y/L$, and

$$\sum_{n\le Y}|a(n)|^2 \le 4\pi e^2 Y\int_R F\,\mathrm{d}\mu \le 4\pi e^2\,Y\Big(1+\frac{54Y}{L}\Big)\|G\|^2,$$

which is (1). For $G=G_{m,m'}$, $L=4mm'$ and $Y\ge mm'$ the bracket is $\le 1+54Y/(4mm') \le 15Y/(mm')$, and Theorem 1.1(i) gives (2). □

Remark 2.1. The argument is the classical one for bounding $\sum_{n\le Y}|a(n)|^2$ by the norm (compare [5, Chapter 5]); its only input is the invariance of $y^2|G|^2$. The strip $y_0\le y\le 2y_0$ with $y_0=1/(4\pi Y)$ is chosen so that the exponential factor in Parseval's identity stays at e^{-2} for all $n\le Y$. For $Y\ge L$ the factor Y^2/L in (1) is of the correct order of magnitude: by the Rankin–Selberg method, $\sum_{n\le Y}|a(n)|^2/n \sim 16\pi^2\|G\|^2 Y/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})$ with $\operatorname{vol} = \frac{\pi}{3}L\prod_{p\mid L}(1+1/p)$ (measure $\mathrm{d}\mu$), so that $\sum_{n\le Y}|a(n)|^2 \sim 8\pi^2\|G\|^2Y^2/\operatorname{vol}$ for fixed G as $Y\to\infty$. Theorem 1.2 makes this uniform in the level, at the cost of the constant.

3. Proof of Theorem 1.3 and Corollary 1.4

3.1. **The remainder on average.** We follow the proof of Theorem 3.2 of Part I, replacing the uniform bound for $|T-M|$ by Theorem 1.2 after an application of Cauchy–Schwarz over N. Recall from Part I, (15), that for $N\equiv 2 \pmod 4$ and coprime square-free d,e with $(de,2N)=1$,

$$R(d,e;N) = \sum_{t\mid N^\flat}\mu(t)\sum_{m\mid d^2}\sum_{m'\mid e^2} c_d(m)c_e(m')\sum_{\substack{t_1\mid t^2\\ t\mid t_1}} c_t(t_1)\,\big(a_G(N)-a_G(N/2)\big), \qquad G=G_{t_1 m,m'}, \tag{4}$$

where $N^\flat$ is the product of the odd primes dividing N, $m = ds$, $m' = es'$ with $s \mid d$, $s' \mid e$, and likewise $t_1 = tu$ with $u \mid t$ (the coefficients $c_d(m)$, $c_t(t_1)$ of Part I, (14), vanish unless $m = ds$ with $s \mid d$, resp. $t_1 = tu$ with $u \mid t$; in particular $t \mid t_1$); the parameters $t_1 m, m'$ of the form $Q_{t_1 m, m'}$ are coprime, odd and cube-free, $\tilde{m}\tilde{m}' := t_1 m m' = t_1 ss' \cdot de$, and the coefficients c satisfy $|c_d(m) c_e(m') c_t(t_1)| \le 3^{\omega(d)+\omega(e)+\omega(t)}$. (For the reader's convenience: c_d is the multiplicative coefficient of Part I, (14), with $c_p(p) = 2$, $c_p(p^2) = -1$ for a split prime p and $c_q(q) = 0$, $c_q(q^2) = 1$ for an inert prime q; it encodes the identities $r(n) = 2r(n/p) - r(n/p^2)$ and $r(n) = r(n/q^2)$, which turn the condition $d \mid n$ into a combination of theta series $\theta^2(mz)$. The sum over t is the Möbius inversion of the condition $(n, N) = 1$, and $a_G(N) - a_G(N/2)$ selects the odd n.) Let $\eta = \varepsilon/10$. For $N \le X$ with $X \ge X_0(\varepsilon)$ we have, as in (F1)–(F3) of Part I: $d_3(de)\, 3^{\omega(d)+\omega(e)+\omega(t)}\, 4^{\omega(N)} d(N)^2 \le X^\eta$; for fixed (d, e, N) the number of tuples (s, s', t, t_1) is at most X^η; and $\sum_{k \le D} d(k)/k \le X^\eta$, $\sum_{k \le D} d(k) k^{-1/2} \le D^{1/2} X^\eta$. We also use the two bounds for a single term,

$$|a_G(N')| \le T(\tilde{m}, \tilde{m}'; N') + M(\tilde{m}, \tilde{m}'; N') \ll X^{2\eta}\Big(\frac{X}{\tilde{m}\tilde{m}'} + 1\Big) \qquad (N' \le X), \tag{5}$$

$$\sum_{n \le X} |a_G(n)| \le X^{1/2}\Big(\sum_{n \le X} |a_G(n)|^2\Big)^{1/2} \ll X\Big(1 + \frac{X}{L}\Big)^{1/2} L^\eta, \tag{6}$$

where $G = G_{\tilde{m}, \tilde{m}'}$ and $L = 4\tilde{m}\tilde{m}'$, the first being (20) of Part I, the second Cauchy–Schwarz, (1) and Theorem 1.1(i); for $L \le X$ the right-hand side of (6) is $\ll X^{3/2} (\tilde{m}\tilde{m}')^{-1/2+\eta}$. Let Σ denote the double sum in Theorem 1.3. Inserting (4) and splitting the tuples according to the size of $t_1 ss'$:

Tuples with $t_1 ss' > X^\varepsilon$. For each N and each tuple, (5) with $\tilde{m}\tilde{m}' > X^\varepsilon de$ gives $|a_G(N')| \ll X^{2\eta}(X^{1-\varepsilon}/(de) + 1)$. Summing over the $\le X^\eta$ tuples, over $N \le X$ and over $de \le D$ with the weights $d_3(de) \cdot 3^\omega \le X^\eta$, these tuples contribute

$$\ll X^{1+4\eta} \sum_{k \le D} d(k)\Big(\frac{X^{1-\varepsilon}}{k} + 1\Big) \ll X^{2-\varepsilon+5\eta} + X^{1+5\eta} D \ll X^{2-\varepsilon/2}.$$

Tuples with $t_1 ss' \le X^\varepsilon$. Now $t \le t_1 \le X^\varepsilon$ (as $t \mid t_1$), so t runs over at most X^ε values, t_1 over at most $d(t^2) \le X^\eta$ values, and (s, s') over at most $d(d) d(e) \le X^\eta$ values; the forms $G = G_{t_1 m, m'}$ have level $L = 4 t_1 ss' de \le 4X^\varepsilon D \le X$, so that the right-hand side of (6) is $\ll X^{3/2} (\tilde{m}\tilde{m}')^{-1/2+\eta}$. For a fixed tuple and fixed (d, e), we drop the condition $t \mid N$ except for the count: the number of $N \le X$ with $t \mid N$ is $\le X/t$, and Cauchy–Schwarz gives

$$\sum_{\substack{N \le X \\ t \mid N}} \big(|a_G(N)| + |a_G(N/2)|\big) \le 2\Big(\frac{X}{t}\Big)^{1/2}\Big(\sum_{n \le X} |a_G(n)|^2\Big)^{1/2}$$

$$\ll \Big(\frac{X}{t}\Big)^{1/2} X\, (\tilde{m}\tilde{m}')^{-1/2+\eta} \le X^{3/2+\eta}\, t^{-1/2} (de)^{-1/2},$$

using (1) in the form $\sum_{n \le X} |a_G(n)|^2 \ll X^2 \|G\|^2 / L$ and $(\tilde{m}\tilde{m}')^{-1/2+\eta} \le X^\eta (de)^{-1/2}$. Summing over $t \le X^\varepsilon$ ($\sum_t t^{-1/2} \le 2X^{\varepsilon/2}$), over the other tuple variables ($\le X^{2\eta}$), and over $de \le D$ with the weights $\le X^\eta$,

$$\ll X^{3/2+4\eta+\varepsilon/2} \sum_{k \le D} d(k) k^{-1/2} \ll X^{3/2+5\eta+\varepsilon/2} D^{1/2} = X^{2-\varepsilon+5\eta} = X^{2-\varepsilon/2},$$

since $D^{1/2} = X^{1/2-3\varepsilon/2}$. Both contributions are $\ll X^{2-\varepsilon/2}$, which proves the first assertion. (The margin is small but positive: the choice $D = X^{1-3\varepsilon}$ saves $X^{-3\varepsilon/2}$ in $D^{1/2}$, of which the sum over t costs $X^{\varepsilon/2}$ and the divisor-type factors cost $X^{5\eta} = X^{\varepsilon/2}$; this is why $\eta = \varepsilon/10$.)

For the second, let $S(N)$ be the inner sum with $D = X^{1-3\varepsilon}$. The number of $N \in (X/2, X]$, $N \equiv 2 \pmod 4$, with $S(N) > X^{1-\varepsilon/4}$ is at most $\Sigma/X^{1-\varepsilon/4} \ll X^{1-\varepsilon/4}$. For the remaining N, since $N^{1-3\varepsilon} \le D$ and $X^{1-\varepsilon/4} \le (2N)^{1-\varepsilon/4} \le N^{1-\varepsilon/5}$ for large X, (3) holds. □

Remark 3.1. The exponent 1 is the balance $X^{3/2}D^{1/2} = X^2$ between the mean-square bound $X^{3/2}(de)^{-1/2}$ for the sum over N of one remainder and the number D of moduli; with a norm bound $\|G_{m,m'}\| \ll (mm')^{\beta+\varepsilon}$ in place of Theorem 1.1(i) one gets the level $1/(2(1+\beta))$ for every N (Part I, Remark 3.3) and $1/(1+2\beta)$ for almost all N; Waibel's $\beta = 1/4$ gives $2/5$ and $2/3$, and $\beta = 0$ gives $1/2$ and 1. The exceptional set $X^{1-\varepsilon/4}$ comes from Markov's inequality applied to a first moment of $S(N)$, which in turn rests on the second moment of $a_G(N)$; a bound for a higher moment would shrink it, and we do not pursue this.

3.2. **The sieve at the level** 19/20**.** Proposition 4.3 of Part I and its proof use the level of distribution only through the bound (3) (there with 2/5); they hold verbatim, for a given N, with $\theta = 2/5$ replaced by any $\theta \in (0,1)$ for which (3) holds with $N^{\theta-\varepsilon}$ in place of $N^{1-3\varepsilon}$, provided $\alpha < \beta_1 < \theta$. Lemmas 4.1 and 4.2 of Part I do not involve the level at all. Hence, for every N satisfying (3) with $3\varepsilon < 1/20$, the conclusion of Proposition 4.3 holds with $\theta = 19/20$, and the proof of Theorem 1.1 of Part I goes through with the parameters certified for $\theta = 19/20$. The script `certify_modular.py` of Part I, extended by the argument `19/20` in the repository of this paper, certifies in interval arithmetic

$$\theta = \tfrac{19}{20}, \qquad \alpha = \tfrac{2}{25}, \qquad \lambda = \tfrac{1}{3}, \qquad \beta_1 = \tfrac{69}{100},$$
$$\tfrac{1}{\lambda} + \tfrac{2}{\beta_1} = \tfrac{407}{69} < 6, \qquad \tfrac{1}{\lambda} + \tfrac{1}{\beta_1} = \tfrac{307}{69} < 5, \qquad L - \lambda R \ge 0.085058$$

(with $L \ge 0.999484$, $R \le 2.743278$, $\theta_A = 0.4740$; the output files are listed at the end of the paper). By Lemma 4.2 of Part I this gives $\Omega(n(N-n)) \le 5$ and $\max(\Omega(n), \Omega(N-n)) \le 4$, and the weight of each n counted is $w(n) = 16 \cdot 2^{\Omega(n(N-n))} \le 2^9$. As in the proof of Theorem 1.1 of Part I, the number of integers counted is at least

$$\big(0.085058 + o(1)\big)\, 2^{-9} \cdot 8\, \mathfrak{S}_r(N)\, c_{2h}\, \frac{N}{\alpha^2 (\log N)^2} \;\ge\; 0.34\, \mathfrak{S}_r(N) \frac{N}{(\log N)^2},$$

since $0.085058 \cdot 2^{-9} \cdot 8 \cdot 1.66485 \cdot (25/2)^2 = 0.3457\ldots$. The exceptional set is that of Theorem 1.3 with, say, $\varepsilon = 1/80$, so $\delta = 1/320$ is admissible. □

Remark 3.2. The same sieve gives $(8,5)$ at the level 1/2 (Part I), $(6,4)$ at 2/3 (certified: $\alpha = 3/50$, $\lambda = 1/3$, $\beta_1 = 51/100$, $L - \lambda R \ge 0.036741$) and $(5,4)$ at 19/20. The floating-point scan of Figure 2 shows that $1/\lambda + 2/\beta_1$ drops below 6 only for $\theta > 0.83$, and that the best attainable value of $1/\lambda + 1/\beta_1$ drops below 4 for $\theta > 0.88$: at $\theta = 19/20$ the minimum is 3.83, attained at $\alpha = 0.10$, $\beta_1 = 0.47$, $\lambda = 0.59$, where $1/\lambda + 2/\beta_1 = 5.96$, so that the Richert weights give $(5,3)$ for almost all N in the floating-point model, with the sifting parameter α at the edge of the scanned range. We have not certified this set, because Corollary 1.4 is in any case superseded for almost all N by the circle method (Section 1); it is an "almost all N" statement with a power-saving exceptional set and says nothing about any individual N, which is where the sieve route matters (Section 4.1).

Remark 3.3 (The vector sieve is not the natural sieve here). The sieve of Part I treats the two sides n and $N-n$ by the vector sieve of Brüdern and Fouvry, with two linear components of levels N^{θ_A} and $N^{\theta-\theta_A}$ and the lower bound $L = f_1(s_A)F_1(s_B) + F_1(s_A)f_1(s_B) - F_1(s_A)F_1(s_B)$. For $s_A, s_B \le 3$ one has $f_1/F_1(s) = \log(s-1)$, so $L > 0$ if and only if $(s_A - 1)(s_B - 1) > e$, and with $s = s_A + s_B$ fixed this requires $s > 2 + 2\sqrt{e} = 5.297\ldots$, attained at the symmetric split. The product sequence $n(N-n)$, however, is a sieve problem of dimension 2 in the classical sense: under the weight $r(n)r(N-n)$ the density of $p \mid n(N-n)$ is $2h(p)$, $\sum_{p<z} 2h(p) = 2\log\log z + O(1)$, the

condition $d \mid n(N-n)$ with $(n,N)=1$ splits uniquely as $d=d_1d_2$, $d_1 \mid n$, $d_2 \mid N-n$, and the remainder for the modulus d is $\sum_{d_1d_2=d} R(d_1,d_2)$, which is exactly what Theorem 3.2 of Part I and Theorem 1.3 bound (this is the reason for the weight $d_3(de)$ there). The two-dimensional sieve of Diamond, Halberstam and Richert [3] applies to it directly, with the joint level $d \le N^\theta$ and the sieving limit $\beta_2 = 4.2665\ldots$ in place of 5.297; its upper and lower functions F_2, f_2 should be used in place of the products of F_1, f_1 both in L and in the weight term R. We expect this to improve the pairs (k,k') at every level without any new analytic input, and we have not carried it out; the bound for $\max(\Omega(n),\Omega(N-n))$ then needs a separate argument, since the two-dimensional sieve controls only $\Omega(n(N-n))$. For comparison, for the unweighted sequence $n(N-n)$, which has level of distribution 1 trivially, the two-dimensional weighted sieve gives $N=P_2+P_3$ for every large even N (Wang Yuan, 1957).

4. The uniform bound and the Poincaré series

Let $L=4mm'$ and let $P_N \in S_2(\Gamma_0(L))$, $N\ge 1$, be the Riesz representative of the N-th coefficient: the unique cusp form with

$$\langle f, P_N\rangle = \frac{a_f(N)}{4\pi N} \qquad \text{for all } f \in S_2(\Gamma_0(L)), \tag{7}$$

i.e. $P_N = (4\pi N)^{-1}\sum_{f\in\mathcal{B}} \overline{a_f(N)}\, f$ for an orthonormal basis $\mathcal{B}$. (This is the N-th Poincaré series; for weight 2 the series itself converges only conditionally, which is why we define P_N by (7).) By (11) of Part I,

$$\|P_N\|^2 = \frac{a_{P_N}(N)}{4\pi N} = \frac{1}{(4\pi N)^2}\sum_{f\in\mathcal{B}} |a_f(N)|^2 = \frac{\Delta_L(N,N)}{4\pi N},$$

where $\Delta_L(N,N)$ is the Petersson density. Hence, for $G=G_{m,m'}$,

$$|a_G(N)|^2 = (4\pi N)^2 |\langle G, P_N\rangle|^2 = 4\pi N\, \Delta_L(N,N)\, \|G\|^2 \cos^2\angle(G,P_N), \tag{8}$$

where $\cos\angle(G,P_N) = |\langle G,P_N\rangle|/(\|G\|\,\|P_N\|)$. The identity separates the three quantities that enter any bound for $a_G(N)$: the norm $\|G\|^2$, which by Theorem 1.1(i) is $\ll L^\varepsilon$; the Petersson density, which is $\le C'$ for $N \le L^{1/2}$ (Part I, Lemma 2.2, Step 4) and $1+O(N^{1/2+\varepsilon}/L)$ in general; and the alignment $\cos^2\angle(G,P_N)\le 1$. Theorem 1.1(ii) is obtained from (8) by bounding the alignment trivially: after the coprime part of N is moved into a Hecke operator (Part I, Lemma 2.2), only P_{n_2} with $n_2 \in \{1,2\}$ occurs, where the density is bounded, and the norm does the rest. Theorem 1.2 says what the alignment is on average: inserting (8) into (1) and restricting to $N \in (Y/2, Y]$,

$$\frac{1}{Y/2}\sum_{Y/2<N\le Y} \Delta_L(N,N)\cos^2\angle(G,P_N) \;\le\; 4e^2\Big(\frac{1}{Y}+\frac{54}{L}\Big) \;\ll\; \frac{1}{L} \qquad (Y\ge L), \tag{9}$$

the value of $\cos^2$ for a unit vector in general position in the $\asymp L$-dimensional space $S_2(\Gamma_0(L))$: on average over N, the theta series is no more aligned with the Poincaré series than a random cusp form of level L would be, and Theorem 1.1(ii) says that no single N is aligned by more than $N^\varepsilon L/\|G\|^2 \gg L^{1-\varepsilon}$ times the average. Expanding in an orthogonal basis $\mathcal{B}^*$ of Hecke eigenforms as in Part I, Section 5.5,

$$\langle G, P_N\rangle = \sum_{\varphi\in\mathcal{B}^*} \frac{\langle G,\varphi\rangle\langle\varphi,P_N\rangle}{\|\varphi\|^2} = \frac{1}{4\pi N}\sum_{\varphi\in\mathcal{B}^*} \frac{\langle\Theta,\varphi\rangle\, a_\varphi(N)}{\|\varphi\|^2},$$

so that Theorem 1.1(ii) is the bound $\ll N^{1/2+\varepsilon}$ for this spectral sum, uniformly in m,m'; the first two versions of Part I asked for a proof through the cancellation between the coefficients $\langle\Theta,\varphi\rangle$ and $a_\varphi(N)$. The proof in Part I does not see this cancellation: it shows instead that

$\sum_\varphi |\langle\Theta,\varphi\rangle|^2/\|\varphi\|^2 = \|G\|^2$ is itself small, i.e. that the theta series has a small cuspidal projection to begin with, and the spectral sum is then bounded by Cauchy–Schwarz.

4.1. **The individual bound.** Theorem 1.2 and the data (Section 5.4) say that the cancellation which the proof does not see is there for most N, and the sieve for every N would profit from it for each N. We state the strongest form suggested by the data.

Conjecture 4.1. For coprime odd m, m' and $N \geq 1$ with $(N, mm') = 1$ and $v_2(N) \leq 1$,

$$|T(m,m';N) - M(m,m';N)| = |a_{G_{m,m'}}(N)| \ \ll_\varepsilon \ N^{1/2+\varepsilon}(mm')^{-1/2+\varepsilon}.$$

In the terms of (8): $\Delta_L(N,N)\cos^2\angle(G,P_N) \ll N^\varepsilon L^{-1+\varepsilon}/\|G\|^2$, i.e. no Poincaré series P_N is aligned with G by more than $N^\varepsilon L^\varepsilon$ times the average (9).

Conjecture 4.1 is the mean-square bound (2) applied to each N separately; it is consistent with the trivial lower bound $|a_G(N)| \geq \operatorname{dist}(M, 8\mathbb{Z})$ for $N \ll L$, where $M \asymp N/(mm') \ll 1$, and with the data of Section 5.4, where the supremum over 10^6 to $4\cdot 10^6$ values of N exceeds the root mean square by a factor between 5 and 10 at every level computed. We do not know how to prove any power saving in mm' for an individual N. In the spectral form the question is a twisted first moment $\sum_\varphi \langle\Theta,\varphi\rangle a_\varphi(N)/\|\varphi\|^2$ over the forms of level L with the "long" twist $N \geq L$ (the sieve at the level θ uses $N \approx L^{1/\theta}$), outside the range $N \leq L^{1-\delta}$ in which the Petersson formula evaluates such moments. Any saving would raise the level for every N:

Proposition 4.2. *Let $0 \leq \eta \leq \frac12$ and suppose that $|a_{G_{m,m'}}(N)| \ll_\varepsilon N^{1/2+\varepsilon}(mm')^{-\eta+\varepsilon}$ for all coprime odd m, m' and all N with $(N, mm') = 1$, $v_2(N) \leq 1$. Then Theorem 3.2 of Part I holds for every $N \equiv 2 \pmod 4$ with $D = N^{\theta-\varepsilon}$, $\theta = 1/(2(1-\eta))$, i.e. $r(n)r(N-n)$ has level of distribution $1/(2(1-\eta))$. In particular, Conjecture 4.1 gives the level $1-\varepsilon$ for every N, and then the sieve of Section 3 gives, for every sufficiently large $N \equiv 2 \pmod 4$, $\Omega(n(N-n)) \leq 5$ and $\max(\Omega(n), \Omega(N-n)) \leq 4$ for at least $0.34\,\mathfrak{S}_r(N)N/(\log N)^2$ integers n.*

Proof. We follow the proof of Theorem 3.2 of Part I and only change the bound for one remainder. Each term of $R(d,e)$ is $\pm c\, a_G(N')$ for a form $G = G_{\tilde m,\tilde m'}$ with $\tilde m\tilde m' = t_1 ss'\cdot de$, where $t \mid N^\flat$, $t_1 \mid t^2$, and $N' \in \{N, N/2\}$ may have common factors with $\tilde m\tilde m'$ at the primes of t only. Lemma 2.5 of Part I reduces $a_G(N')$ to $\leq 4^{\omega(N)}d(N)$ terms $a_{G_{m_j,m_j'}}(N_j)$ with $(N_j, m_jm_j') = 1$, $N_j \mid N'$ and $m_jm_j' \geq \tilde m\tilde m'/t_1^2 \geq de/t_1^2$ (each reduction step at a prime p divides the product of the coefficients by at most p^2, and only the primes of t_1 are involved). Hence, by the hypothesis with $\varepsilon/20$ in place of ε,

$$|a_G(N')| \ll 4^{\omega(N)}d(N)\, N^{1/2+\varepsilon/20}\,(de)^{-\eta}\, t_1^{2\eta}\,(mm')^{\varepsilon/20}.$$

Put $\eta' = \varepsilon/20$. For the terms with $t_1ss' \leq N^\varepsilon$ this is $\ll N^{1/2+3\eta'+2\varepsilon\eta}(de)^{-\eta}$ by (F1) of Part I, and their total is

$$\ll N^{1/2+5\eta'+\varepsilon}\sum_{k\leq D} d(k)k^{-\eta} \ll N^{1/2+6\eta'+\varepsilon}D^{1-\eta} = N^{1/2+6\eta'+\varepsilon}N^{1/2-\varepsilon(1-\eta)} \leq N^{1-\varepsilon/2+6\eta'},$$

since $D^{1-\eta} = N^{(\theta-\varepsilon)(1-\eta)} = N^{1/2-\varepsilon(1-\eta)}$ and $1-\eta \geq \frac12$. The terms with $t_1ss' > N^\varepsilon$ are bounded by the trivial bound (5) exactly as in Part I, by $N^{1-\varepsilon+5\eta'} + N^{\theta+5\eta'}$. With $\theta \leq 1-\varepsilon$ (replace ε by $\min(\varepsilon, 1-\theta)$ if necessary) the total is $\ll N^{1-\varepsilon/2+6\eta'} \leq N^{1-\varepsilon/5}$, which is the conclusion of Theorem 3.2 of Part I with $N^{1-\varepsilon/5}$ in place of $N^{1-\varepsilon/3}$; Proposition 4.3 of Part I needs only a power saving. The last assertion is Section 3, where the parameters at the level 19/20 were certified for any N satisfying (3). □

The exponent $\theta = 1/(2(1-\eta))$ interpolates between the level $1/2$ of Part I ($\eta = 0$) and the level 1 ($\eta = \frac{1}{2}$); the level $2/3$, which gives $(6, 4)$ for every N with the certified parameters of Section 3, needs only $\eta = \frac{1}{4}$, i.e. a saving of $(mm')^{-1/4}$ over Theorem 1.1(ii), half of what the mean square (2) gives on average.

5. Numerical evidence

5.1. **The mean square and the norm.** The script `mean_square_check.py` computes, with the tables and the FFT convolution of `sup_error_scan.py` of Part I, the sums $S_2(Y) = \sum_{N\in\mathcal{N}_Y} |a_G(N)|^2$ over the admissible set

$$\mathcal{N}_Y = \{N \le Y : \ (N, mm') = 1, \ v_2(N) \le 3\},$$

on which the main term of that script is exact, for $Y = 2^{21}$ and $Y = 2^{20}$, and the estimate

$$\|G\|^2_{\mathrm{est}} = \frac{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})}{16\pi^2} \cdot \frac{1}{\#\mathcal{N}_Y} \sum_{N\in\mathcal{N}_Y} \frac{|a_G(N)|^2}{N}$$

suggested by the Rankin–Selberg asymptotic of Remark 2.1. Replacing $1/Y$ by $1/\#\mathcal{N}_Y$ assumes that the mean square of $|a_G(N)|^2/N$ over the excluded N (those with $(N, mm') > 1$ or $v_2(N) \ge 4$) is the same as over $\mathcal{N}_Y$; this is immaterial when mm' has only large prime factors, but introduces a bias of unknown sign for $(1, 5)$, $(9, 1)$, $(3, 5)$ and $(1, 15015)$, where 25%, 38%, 50% and 64% of the N are excluded. Table 1 shows the result at $Y = 2^{21}$, and Figure 1 the dependence on Y and on mm'.

(m, m')	mm'	$\#\mathcal{N}_Y$	$S_2(Y)/S_2(Y/2)$	$\|G\|^2_{\mathrm{est}}$	$\|G\|^2_{\mathrm{est}}/(mm')^{1/2}$	$\|G\|^2_{\mathrm{est}}/(\log mm')^2$
$(1, 5)$	5	1572864	4.001	0.74	0.33	0.29
$(9, 1)$	9	1310720	4.001	1.47	0.49	0.30
$(3, 5)$	15	1048576	4.001	3.44	0.89	0.47
$(1, 101)$	101	1946614	4.000	7.95	0.79	0.373
$(13, 29)$	377	1752262	4.001	13.09	0.67	0.372
$(1, 1009)$	1009	1964131	3.976	17.96	0.57	0.375
$(61, 67)$	4087	1904986	3.961	24.47	0.38	0.354
$(101, 103)$	10403	1927715	3.948	29.81	0.29	0.348
$(1, 15015)$	15015	754220	4.359	36.62	0.30	0.396
$(1009, 1013)$	1022117	1962192	4.559	46.87	0.046	0.244

Table 1. Mean square of the cuspidal error over the admissible $N \le Y = 2^{21}$ (`results/mean_square_check_x21.txt`). The ratio $S_2(Y)/S_2(Y/2) \approx 4$ is the growth Y^2 of Theorem 1.2. For the first three pairs and for $(1, 15015)$ a large proportion of the N (25% to 64%) is excluded by the coprimality condition, and for the last pair $Y < L$, so that neither (1) in its Y^2/L form nor the Rankin–Selberg asymptotic applies; these rows are shown for completeness. The exact value for $(1, 5)$ is 0.5785 (Section 5.2).

5.2. **An exact check at the level** 20**.** The space $S_2(\Gamma_0(20))$ is one-dimensional, spanned by the newform $f_{20} = \eta(2z)^2\eta(10z)^2$ of the elliptic curve $20a1$: $y^2 = x^3 + x^2 + 4x + 4$, which is $X_0(20)$ itself (modular degree 1, Manin constant 1). Hence $G_{1,5} = c\, f_{20}$ with $c = a_G(1) = T(1, 5; 1) - M(1, 5; 1) = 4 - \frac{4}{3} = \frac{8}{3}$ (the Siegel product at $n = 1$ is $\frac{\pi^2}{5} \cdot \frac{25}{3\pi^2} \cdot \frac{4}{5}$), and $\|G_{1,5}\|^2 = \frac{64}{9}\|f_{20}\|^2$. The script `norm_level20.py` computes $\|f_{20}\|^2$ in two independent ways: from the Rankin–Selberg asymptotic $\sum_{n\le y} a_f(n)^2/n \sim 16\pi^2\|f\|^2 y/\,\mathrm{vol}$, $\mathrm{vol}(\Gamma_0(20)\backslash\mathbb{H}) = 12\pi$, with the coefficients of the

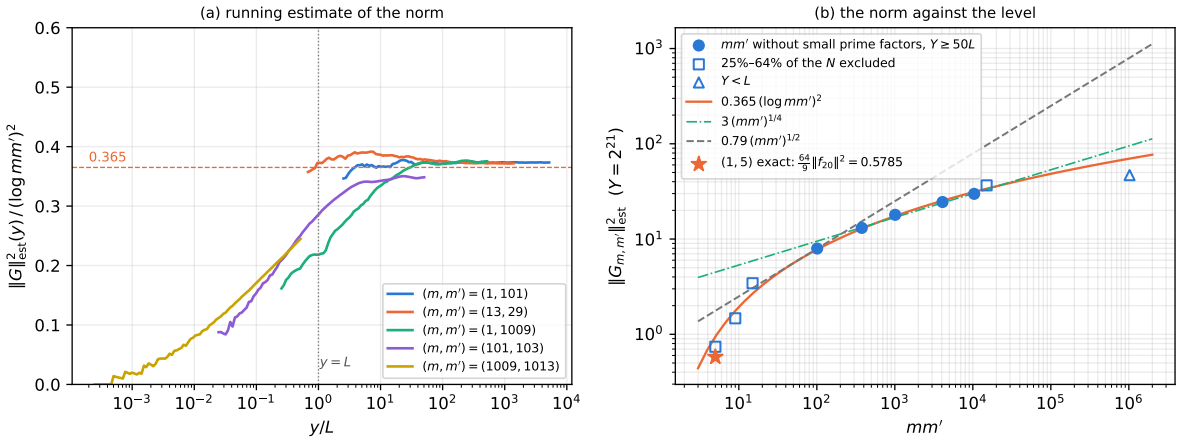


FIGURE 1. (a) The running estimate $\|G\|^2_{\mathrm{est}}(y)$, computed over the admissible $N \le y$, divided by $(\log mm')^2$, against y/L on a logarithmic scale, for five pairs (m, m'); the plateau for $y \gg L$ is the growth Y^2 of Theorem 1.2 and the Rankin–Selberg asymptotic of Remark 2.1, and its height is the constant 0.365 of Section 5.3 (dashed). (b) $\|G_{m,m'}\|^2_{\mathrm{est}}$ at $Y = 2^{21}$ against mm', both axes logarithmic; filled circles: the five pairs with mm' free of small prime factors and $Y \ge 50L$; open squares: pairs with 25%–64% of the N excluded; triangle: $(1009, 1013)$, for which $Y < L$; star: the exact value at $(1, 5)$. The lines are $0.365(\log mm')^2$, the fit of Section 5.3, $3(mm')^{1/4}$, and $0.79(mm')^{1/2}$ (the growth allowed by Waibel's general bound, through the point $(1, 101)$). Script `plot_mean_square.py`; floating point.

eta product up to 2^{20}, which gives 0.08128, 0.08128, 0.08134 at $y = 2^{16}, 2^{18}, 2^{20}$; and from the period lattice Λ of $20a1$, since $\varphi^*\omega = 2\pi i f(z)\mathrm{d}z$ gives $\|f\|^2 = \mathrm{covol}(\Lambda)/(4\pi^2)$ for a modular parametrisation of degree 1. The periods $\omega_1 = 2.824375\ldots$ (real) and $\omega_2 = 2.274165\ldots i$ of $y^2 = (x+1)(x^2+4)$ are computed by quadrature, the lattice $\mathbb{Z}\omega_1 + \mathbb{Z}(\omega_1+\omega_2)/2$ is identified as the period lattice by recomputing $g_2 = -44/3$ and $|g_3| = 296/27$ from it through E_4 and E_6, and its covolume $3.2115478\ldots$ gives $\|f_{20}\|^2 = 0.0813495\ldots$. The two values agree to four decimals. The script also checks that $T(1,5;N) - M(1,5;N)$ as computed by the pipeline of Part I equals $\frac{8}{3}a_f(N)$ on every admissible $N \le 2^{20}$, to 10^{-9}. This validates the Siegel main term, the FFT computation, the Rankin–Selberg normalisation and the volume used in the estimate $\|G\|^2_{\mathrm{est}}$. The exact value $\|G_{1,5}\|^2 = 0.5785$ is below the estimate 0.74 of Table 1, as expected: the excluded N are either multiples of 5, where $|a_f(5n)| = |a_f(n)|$, or divisible by 16, where $a_f = 0$, so the admissible mean is biased upwards.

5.3. **The empirical law.** The last column of Table 1 shows $\|G\|^2_{\mathrm{est}}/(\log mm')^2$. For the five pairs with $10^2 \le mm' \le 1.1 \cdot 10^4$, $Y \ge 50L$ and mm' free of small prime factors, it lies between 0.348 and 0.375, with mean 0.365, while the ratio to $(mm')^{1/2}$ falls from 0.79 to 0.29 and the ratio to $(mm')^{1/4}$ rises from 2.5 to 3.2 over the same range. Over a range of a factor 100 in mm' the fits $(\log mm')^2$ and $(mm')^{1/4}$ are not separated by the data alone, but Theorem 1.1(i) excludes any power, and the heuristic of Part I, Remark 2.13, that $\|G\|^2$ is a sum over the cusps of $\sum_{n \ll mm'} |b(n)|^2/n$ with $|b(n)| \approx r(\alpha)r(\beta)$ on $O(1)$ pairs, predicts $(\log mm')^2$ with a constant of the order of $\frac{1}{4\pi} \cdot 2 \cdot (\text{number of cusp classes with a large contribution}) \approx 0.3$–$0.5$. We therefore

record

$$\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2 \qquad (10^2 \le mm' \le 1.6\cdot 10^4),$$

and, for the root mean square of $(T-M)/\sqrt{N}$ over $N \le Y$, $Y \gg L$,

$$\mathrm{rms}^2 \approx \frac{16\pi^2\|G\|^2}{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})} \approx \frac{44\,(\log mm')^2}{mm'\prod_{p\mid 2mm'}(1+1/p)},$$

which reproduces the column "rms" of Table 3 of Part I (0.27 for $(101,103)$, 0.67 for $(1,1009)$). This corrects the remark in Section 5.3 of the first two versions of Part I that the root mean square "decreases like $(mm')^{-1/4}$" and that Waibel's bound "appears to be sharp": the decrease is like $(mm')^{-1/2}$ up to a logarithm, as Theorem 1.2 with Theorem 1.1(i) shows; version 3 of Part I states the corrected law.

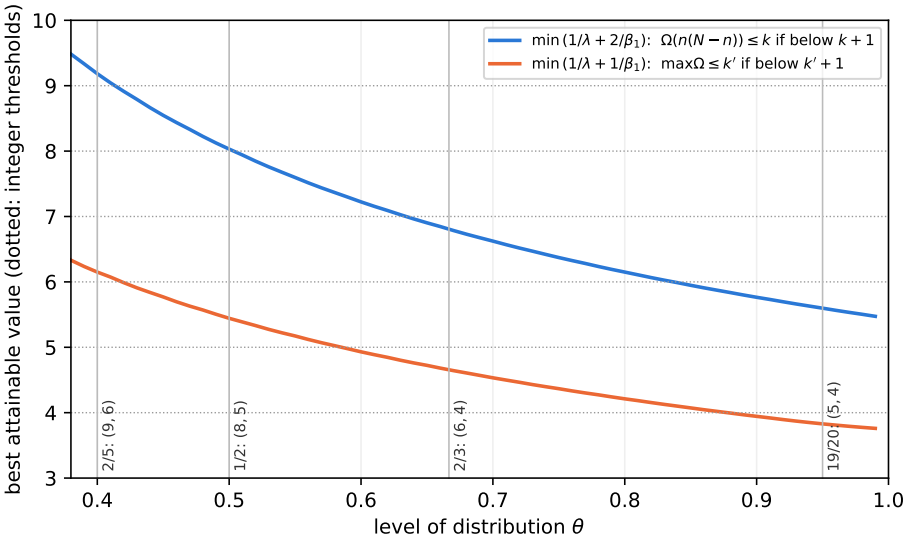


FIGURE 2. The smallest values of $1/\lambda + 2/\beta_1$ and of $1/\lambda + 1/\beta_1$ attainable with $L - \lambda R > 0$ in the sieve of Part I, as functions of the level of distribution θ (floating-point model of `modular_sieve_scan.py`, minimised over α, β_1 with $\lambda = L/R$; script `plot_sieve_thresholds.py`). The bound $\Omega(n(N-n)) \le k$, resp. $\max\Omega \le k'$, holds when the blue, resp. orange, curve is below the dotted line $k+1$, resp. $k'+1$. The vertical lines mark $\theta = 2/5$ (Waibel's bound alone) and $1/2$ (Part I, every N), $2/3$ and $19/20$ (almost all N); the parameter sets at these four levels are certified.

5.4. **The supremum over N.** Conjecture 4.1 says that the supremum of $|a_G(N)|/\sqrt{N}$ over N in a range is, up to $N^{\varepsilon}L^{\varepsilon}$, of the size of the root mean square $\approx 2.7(\log mm')(mm')^{-1/2}$ of Section 5.3. Table 2, extracted from the scans of Part I (`results/sup_error_scan_x21.txt`, `_x23.txt` there), gives the ratio of the supremum over all admissible $N \in (X/2, X]$ to the root mean square, for $X = 2^{21}$ (about 10^6 values of N per pair) and $X = 2^{23}$ (about $4\cdot 10^6$). For the pairs with $L \le 6\cdot 10^4$ the ratio lies between 5.4 and 9.8 and shows no trend in mm'; its mild growth from $X = 2^{21}$ to 2^{23} is the growth of the extreme value of a fixed heavy-tailed distribution with the sample size (the suprema are attained at N with 16–192 divisors, in line with the factor $d(N)$ of Deligne's bound). The pair $(9,1)$ is special: $S_2(\Gamma_0(36))$ is spanned by the newform of the curve $36a1$, $y^2 = x^3 + 1$, which has complex multiplication, so that $a_f(p) = 0$

for $p \equiv 2 \pmod 3$ and the distribution of $a_f(N)$ is far from the Gaussian bulk of Figure 2 of Part I. The pair $(1009, 1013)$ is in the range $X < 2L$, where neither the mean square nor the supremum has reached its asymptotic regime.

(m, m')	mm'	rms	sup, $X = 2^{21}$	sup /rms	sup, $X = 2^{23}$	sup /rms
$(1, 5)$	5	1.76	14.7	8.4	17.3	9.8
$(3, 5)$	15	1.90	10.5	5.5	12.2	6.4
$(1, 101)$	101	1.40	11.8	8.4	12.1	8.6
$(13, 29)$	377	0.89	5.9	6.6	6.8	7.7
$(1, 1009)$	1009	0.67	4.4	6.5	5.3	8.0
$(61, 67)$	4087	0.38	2.5	6.5	2.4	6.3
$(101, 103)$	10403	0.27	1.9	7.3	1.8	6.7
$(1, 15015)$	15015	0.18	1.0	5.7	–	–
$(9, 1)$	9	1.76	28.7	16.3	33.8	19.2
$(1009, 1013)$	1022117	0.036	0.61	17	0.55	14

TABLE 2. The supremum of $|T(m, m'; N) - M(m, m'; N)|/\sqrt{N}$ over the admissible $N \in (X/2, X]$ of Part I, Section 5, and its ratio to the root mean square over the same N (the rms is the same to two decimals for both X). The last two pairs are the degenerate cases discussed in the text.

DATA AND CODE

The scripts `mean_square_check.py`, `norm_level20.py`, `certify_modular.py` (arguments `1/2`, `2/3`, `19/20`), `plot_mean_square.py` and `plot_sieve_thresholds.py` (Figures 1 and 2), the modules of Part I they use (`sup_error_scan.py`, `modular_sieve_scan.py`, `sieve_core.py`) and the outputs `results/mean_square_check_x21.txt`, `results/norm_level20.txt`, `results/certify_modular_output_twothirds.txt`, `results/certify_modular_output_nineteen.txt`, `results/sieve_thresholds.txt` are in the repository https://github.com/Ruqing1963/two-squares-mean-square; the code and data of Part I are at https://github.com/Ruqing1963/two-squares-theta-level. The certification of the sieve parameters uses directed rounding as described in Part I, Appendix A; the computations of Section 5 are in floating point and are evidence, not proof.

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